

2015 Notes from the Marking Centre – Chemistry

Introduction

This document has been produced for the teachers and candidates of the Stage 6 Chemistry course. It contains comments on candidate responses to the 2015 Higher School Certificate examination, highlighting their strengths in particular parts of the examination and indicating where candidates need to improve.

This document should be read along with:

- the [Stage 6 Chemistry syllabus \(PDF\)](#)
- the [2015 Higher School Certificate Chemistry examination \(PDF\)](#)
- the [marking guidelines \(PDF\)](#)
- [advice for candidates attempting science examinations](#)
- [Advice for HSC students about examinations](#)
- other support documents developed by the Board of Studies, Teaching and Educational Standards NSW to assist in the teaching and learning of Chemistry in Stage 6.

Section I

Part B

Candidates showed strength in these areas:

- preparing an indicator (Q21a)
- plotting points on a graph (Q22a)
- outlining the structure of an electrochemical cell (Q23)
- describing the steps of addition polymerisation and relating the use of polymers to their structures and properties (Q25a and b)
- using the correct number of significant figures and units in a calculation (Q26c).

Candidates need to improve in these areas:

- drawing a line of best fit through plotted points (Q22a)
- writing equations to represent the basic nature of the acetate ion and demonstrating how a strong base affects a buffer solution (Q24a and b)
- drawing full structural formulae showing all bonds, including the O-H bond (Q26a)
- identifying the equivalence point of a titration (Q26b)
- discussing the use of a radioisotope in a non-medical industry (Q27)
- justifying the modifications made to the procedure used to collect barium sulphate precipitate (Q29a)
- clearly relating the given graph to the conditions used in the Haber process (Q30).

Section II

Question 31 – Industrial Chemistry

Candidates showed strength in these areas:

- writing a procedure for the safe making of soap (Q31bi)
- using Le Châtelier's Principle to account for changes in concentration (Q31ai)
- explaining the emulsifying properties of soap (Q31bii)
- extracting data from a flow chart (Q31ci).

Candidates need to improve in these areas:

- calculating an equilibrium constant by using an ICE table or similar to determine the equilibrium concentrations (Q31aii)
- identifying the raw materials required for the Solvay process and justifying the location of a Solvay plant (Q31cii)
- comparing the chemistry of two methods of producing sodium hydroxide (Q31d).

Question 32 – Shipwrecks, Corrosion and Conservation

Candidates showed strength in these areas:

- outlining the limitations of using paint to protect ships (Q32ai)
- drawing an electrolytic cell (Q32ci).

Candidates need to improve in these areas:

- describing a valid and reliable procedure that can be used to compare the corrosion rates of iron and a named form of steel (Q32bi)
- identifying two types of steel and relating their percentage composition to their properties (Q32bii)
- identifying the differences in conditions between shallow and deep water and explaining the different rusting processes with equations (Q32d).

Question 33 – The Biochemistry of Movement

Candidates showed strength in these areas:

- identifying components of a cell involved in respiration (Q33ai)
- understanding the function of ATP (Q33aii)
- identifying bonding types and the relationship to protein shape (Q33ci)
- comparing metabolic pathways involved in sprinting and walking (Q33d).

Candidates need to improve in these areas:

- understanding the basic structure of ATP and relating it to biological significance (Q33aii)
- understanding validity and reliability in relation to experimentation (Q33bi)
- recognising the limitations of using models in understanding how enzymes function in living systems (Q33bii)
- relating denaturation of enzymes to their biological activity (Q33cii).

Question 34 – The Chemistry of Art

Candidates showed strength in these areas:

- identifying the chemical composition of cosmetics used in an ancient culture and their potential health risks (Q34ai)
- explaining the emitted flame colour of metal ions after undergoing excitation (Q34bii)
- assessing the contribution of the Bohr model to our understanding of the atomic structure with reference to emission spectra (Q34d).

Candidates need to improve in these areas:

- understanding the formation of complex compounds with ligands and their resulting reflected colours (Q34aii)
- understanding the use of the first ionisation energy to show the existence of sub-shell and using Hund's rule to show the possibility of electron repulsion when two electrons occupy the same orbital (Q34ci).

Question 35 – Forensic Chemistry

Candidates showed strength in these areas:

- extracting information from DNA profile schematics (Q35ci)
- identifying the composition and structure of cellulose (Q35ai)
- describing a test for protein in egg white (Q35bii).

Candidates need to improve in these areas:

- assessing the use of a chromatography technique in the analysis of forensic evidence (Q35d)
- describing a valid procedure (Q35bii)

describing the relationship between amino acids and